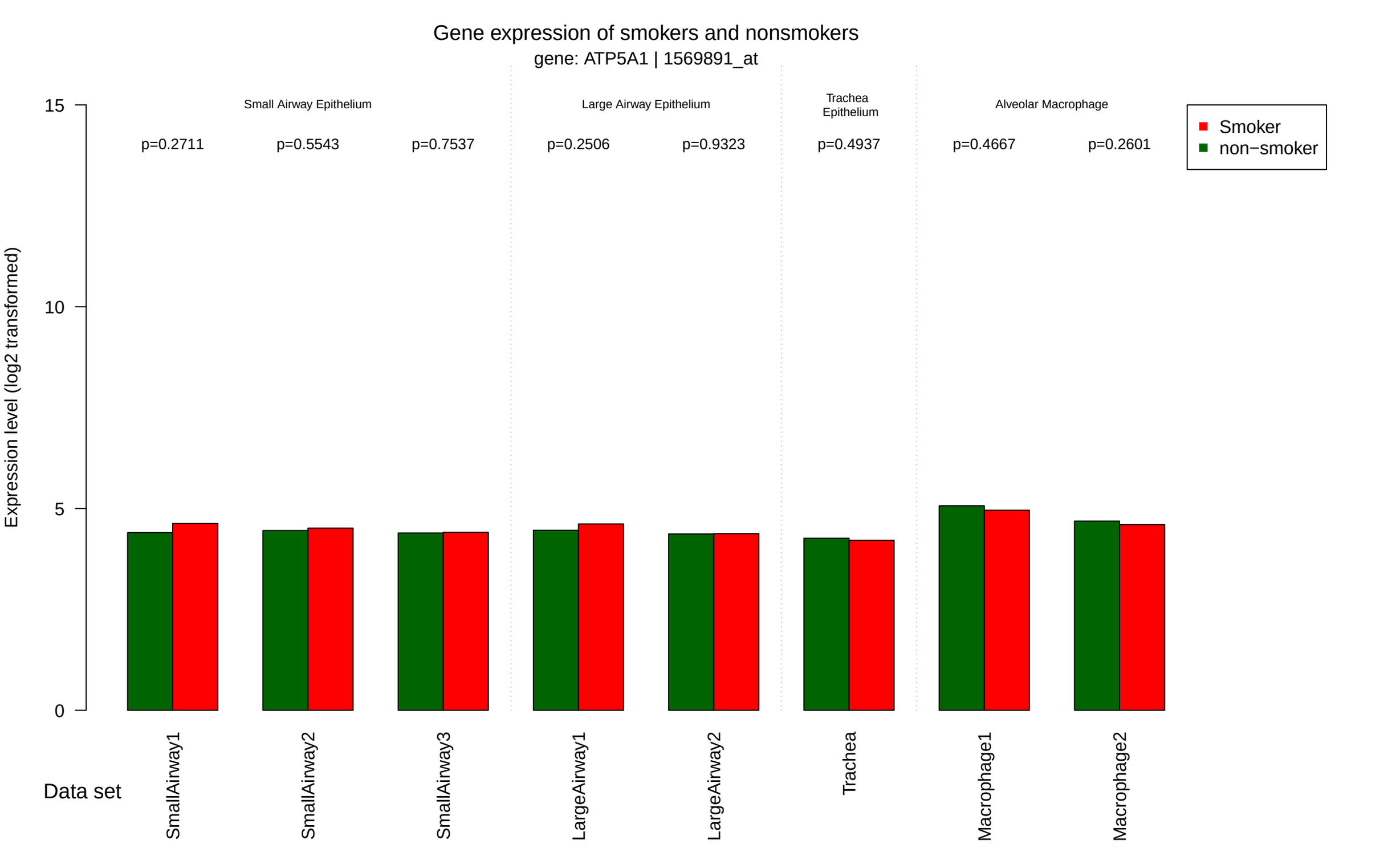
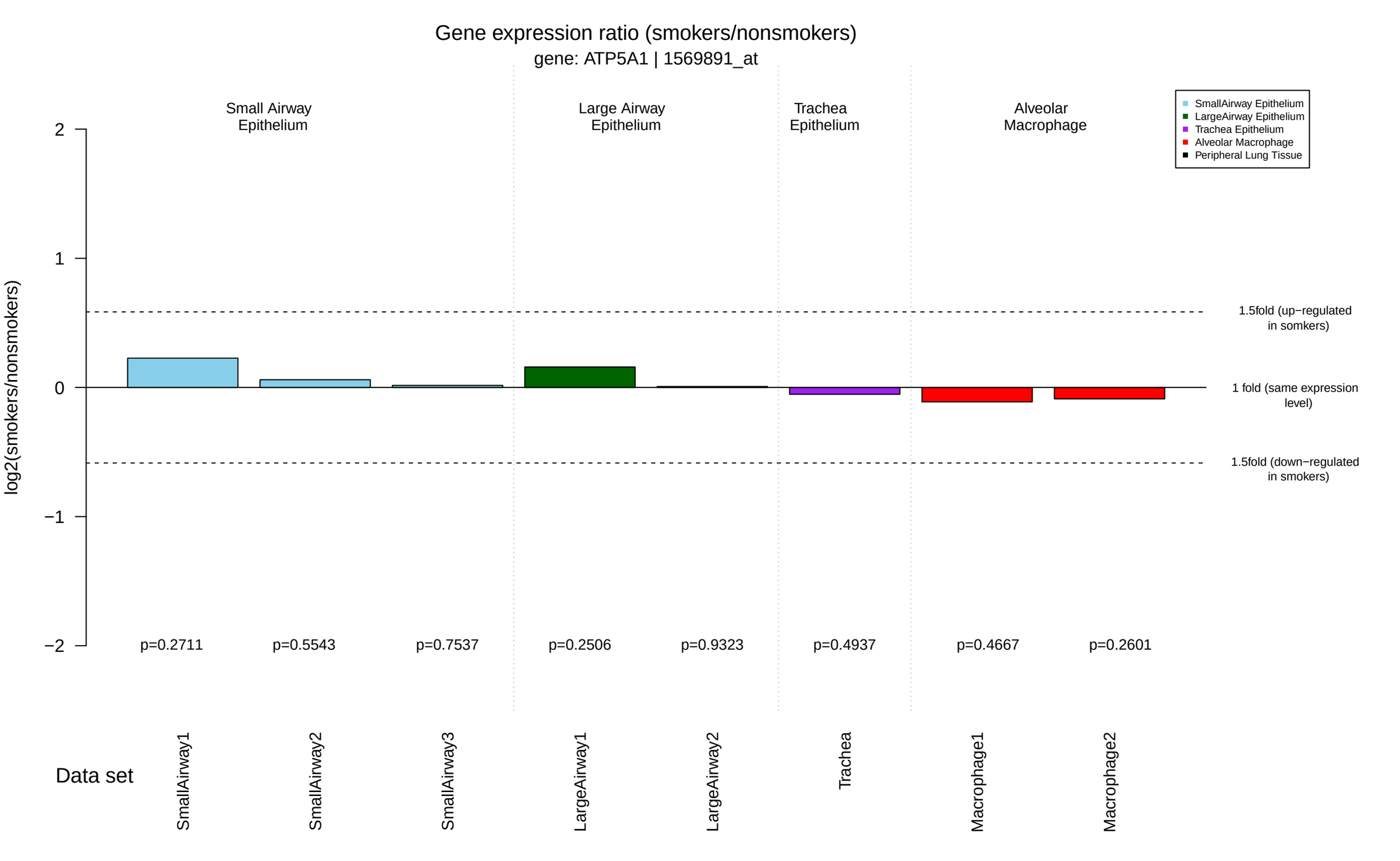
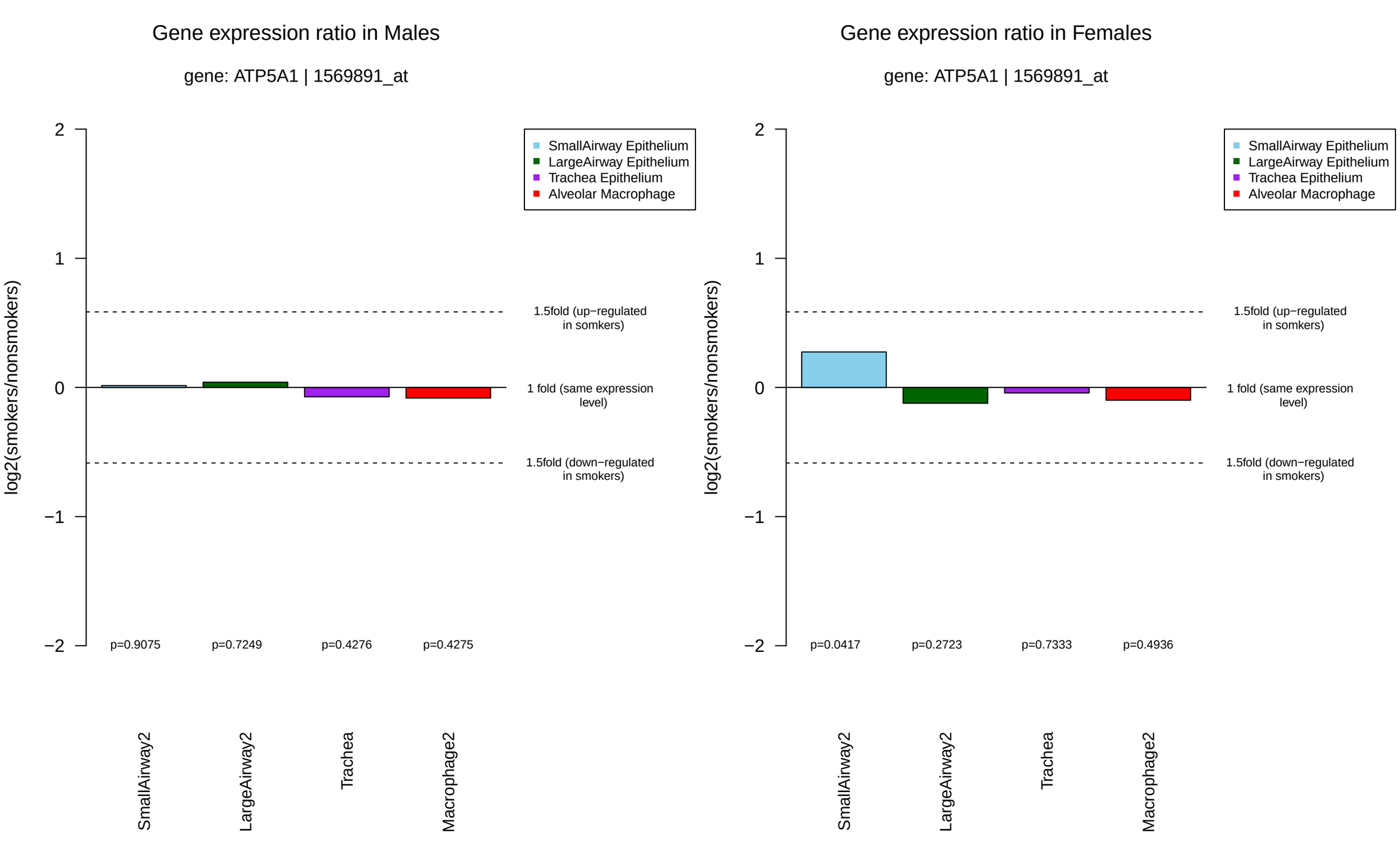
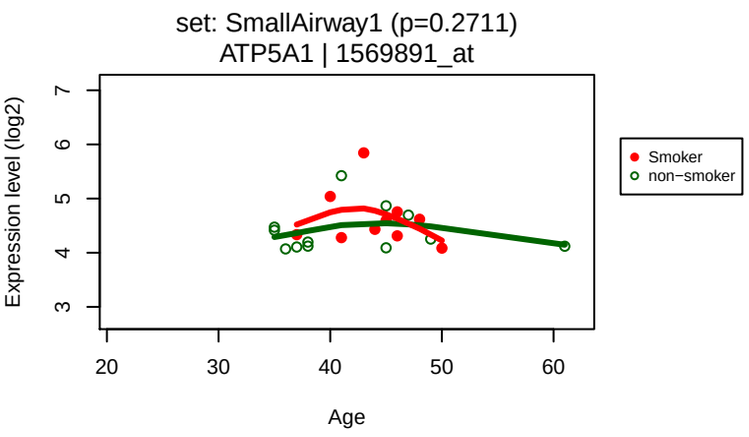
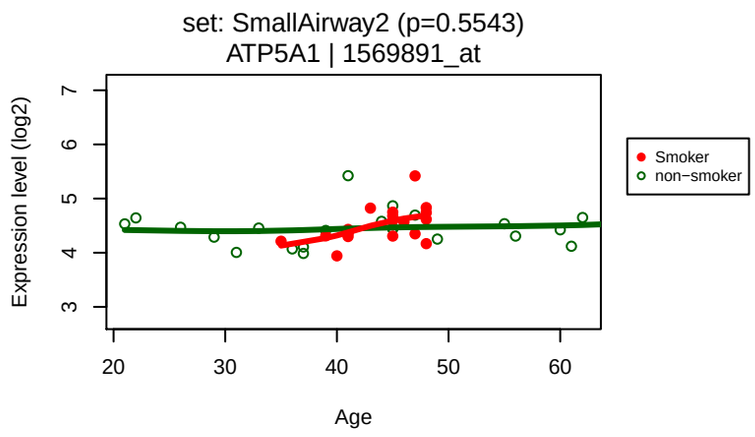
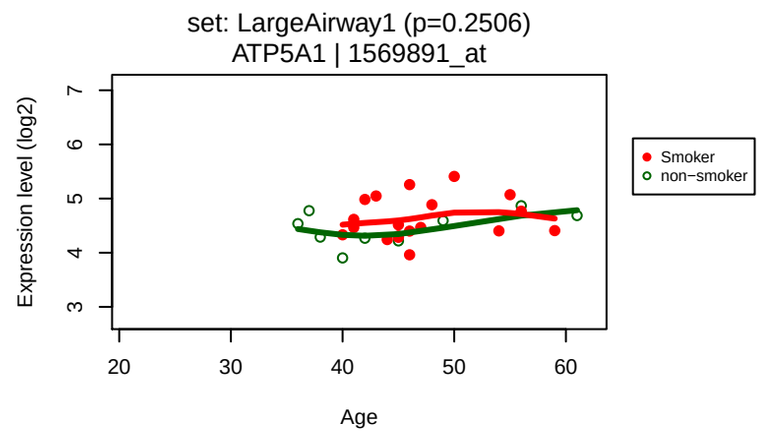

Gene expression of smokers and nonsmokers
gene: ATP5A1 | 1569891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2711
p=0.5543
p=0.7537
p=0.2506
p=0.9323
p=0.4937
p=0.4667
p=0.2601
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

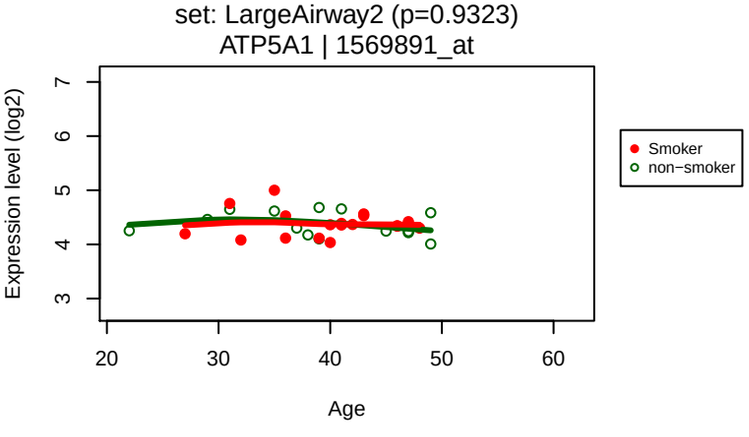

Gene expression ratio (smokers/nonsmokers)
gene: ATP5A1 | 1569891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2711
p=0.5543
p=0.7537
p=0.2506
p=0.9323
p=0.4937
p=0.4667
p=0.2601
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

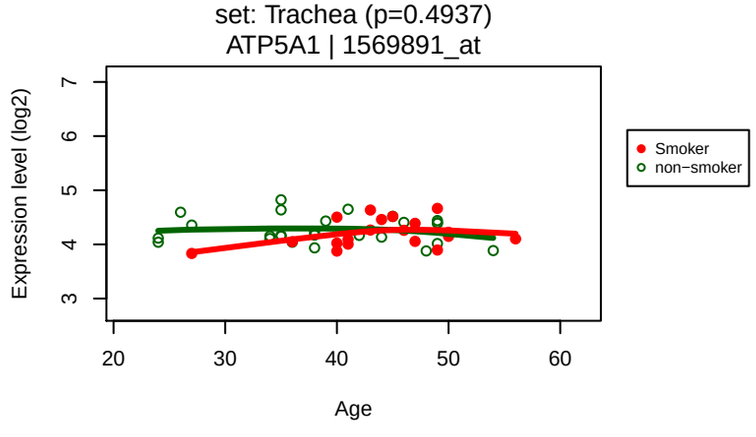

Gene expression ratio in Males
gene: ATP5A1 | 1569891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9075
p=0.7249
p=0.4276
p=0.4275
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP5A1 | 1569891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.2723
p=0.7333
p=0.4936
SmallAirway2
LargeAirway2
Trachea
Macrophage2

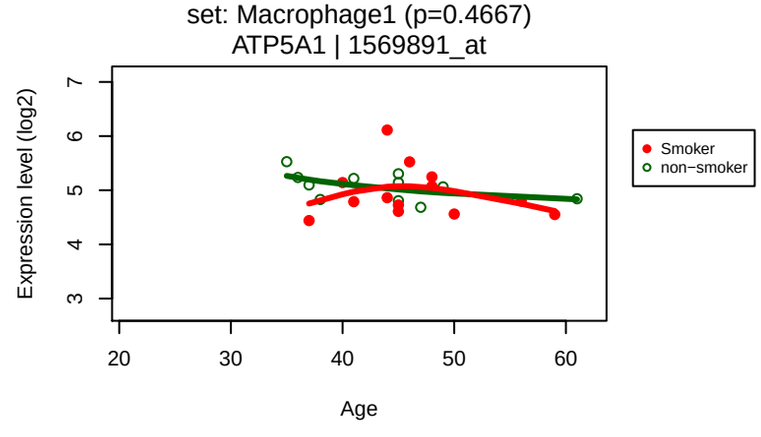

set: SmallAirway1 (p=0.2711)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker

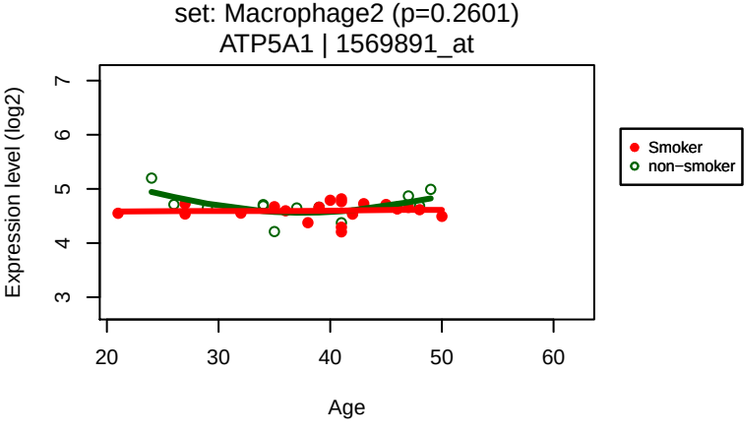

set: SmallAirway2 (p=0.5543)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker

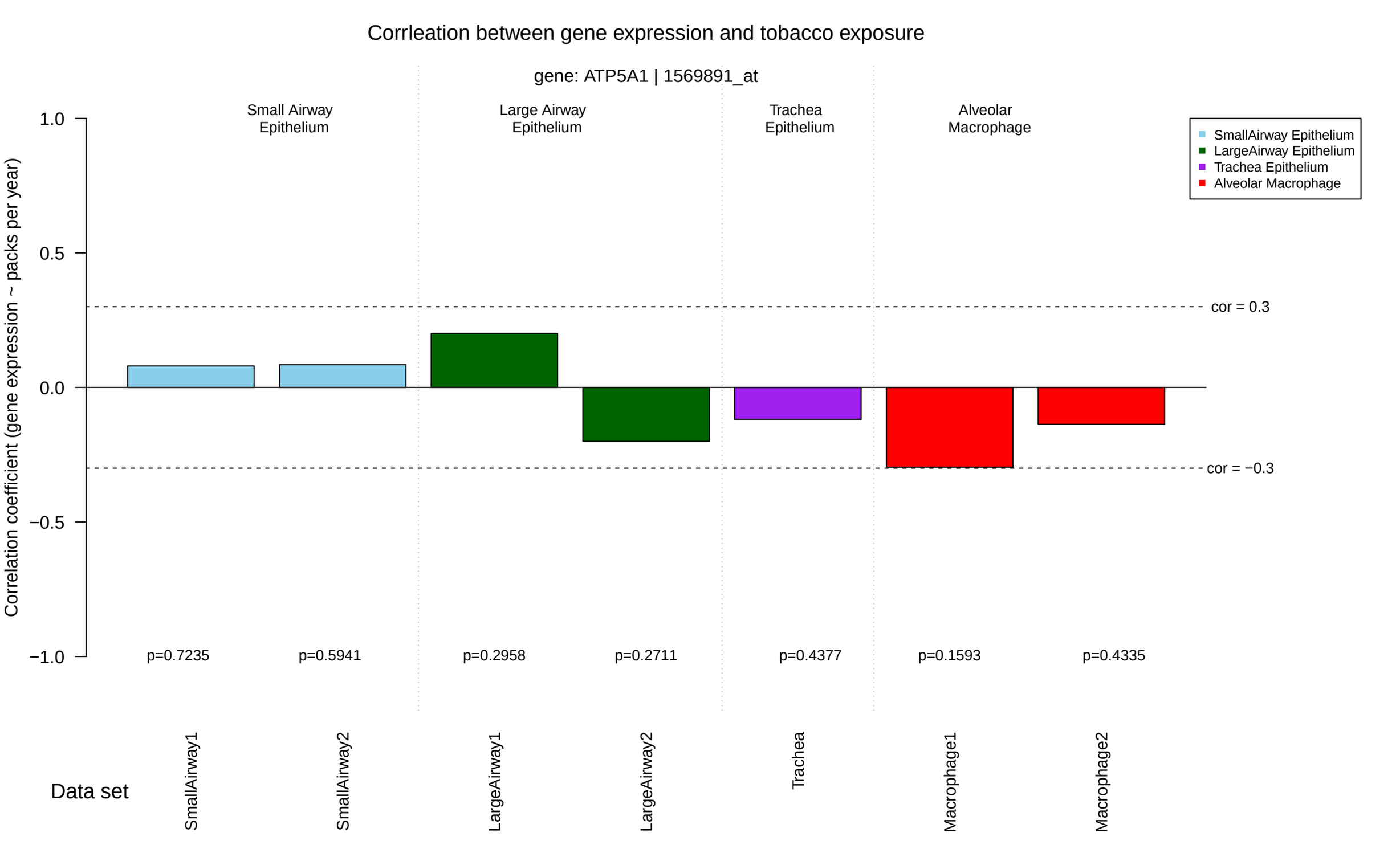

set: LargeAirway1 (p=0.2506)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9323)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4937)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4667)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2601)
ATP5A1 | 1569891_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP5A1 | 1569891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7235
p=0.5941
p=0.2958
p=0.2711
p=0.4377
p=0.1593
p=0.4335
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2